Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: 20260206_171419_resnet50_classifier.pt

Attack Type: FGSM
Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
100.0%	35.35%	34.82%	5.30s
			Accuracy drop: 0.53%

Threat Level Assessment

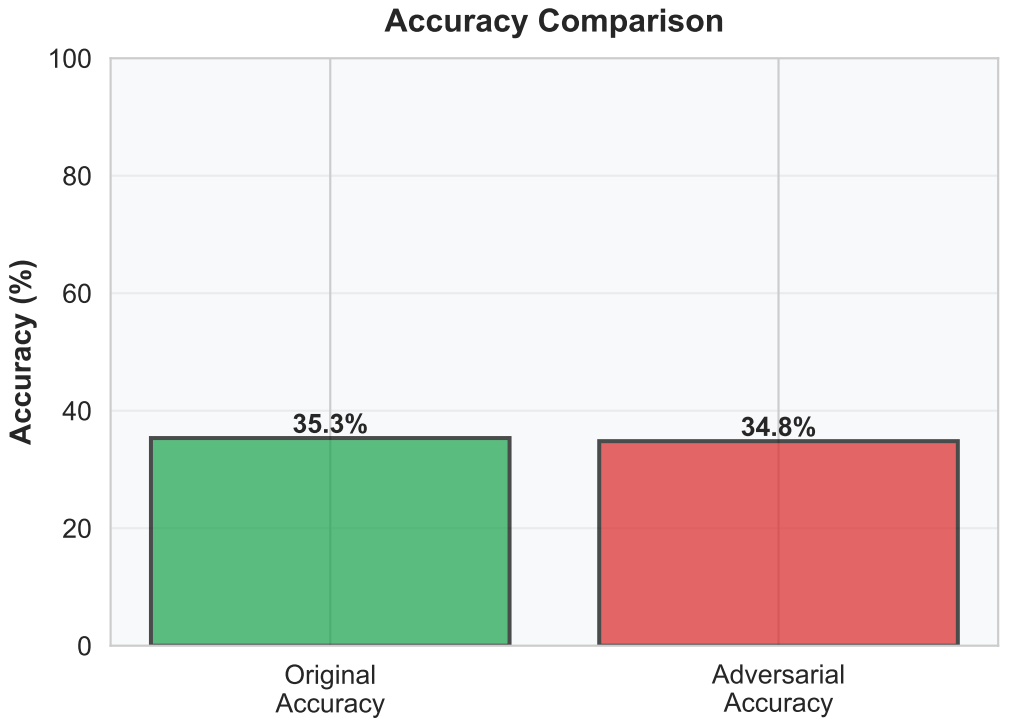
HIGH RISK

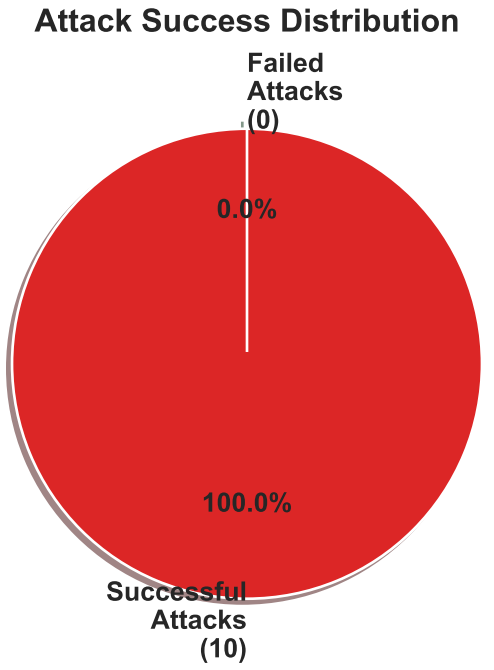
The FGSM attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 35.35% to 34.82%, indicating the current robustness posture is high risk.

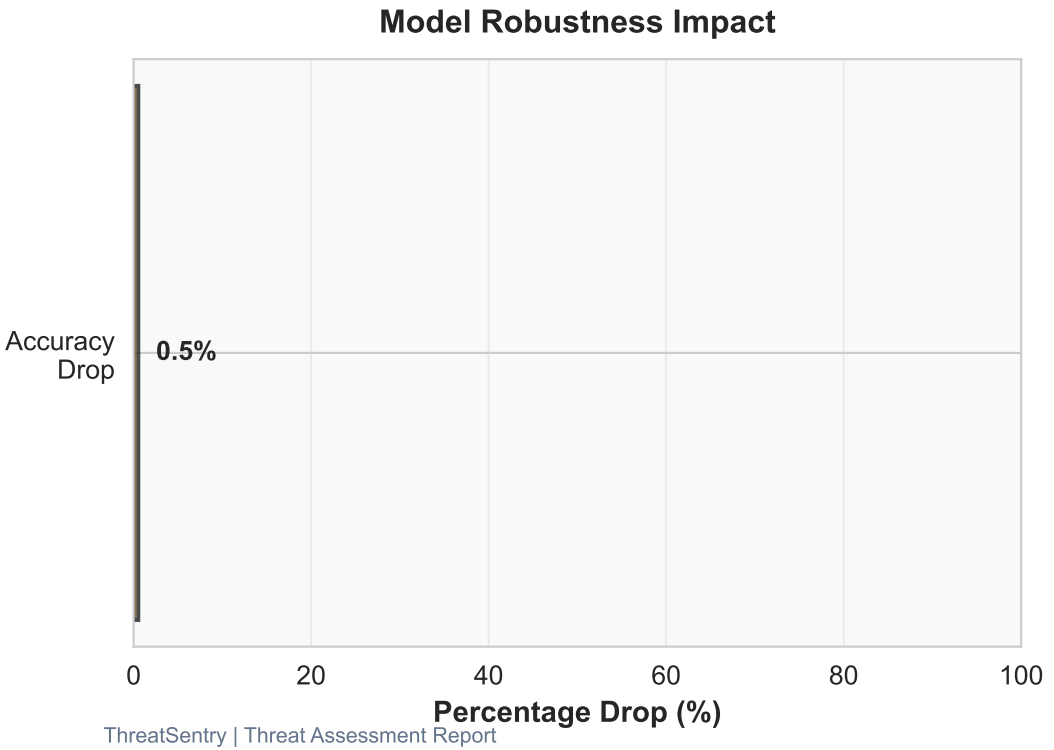
Assessment Narrative

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 35.35%, Average adversarial accuracy: 34.82%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis







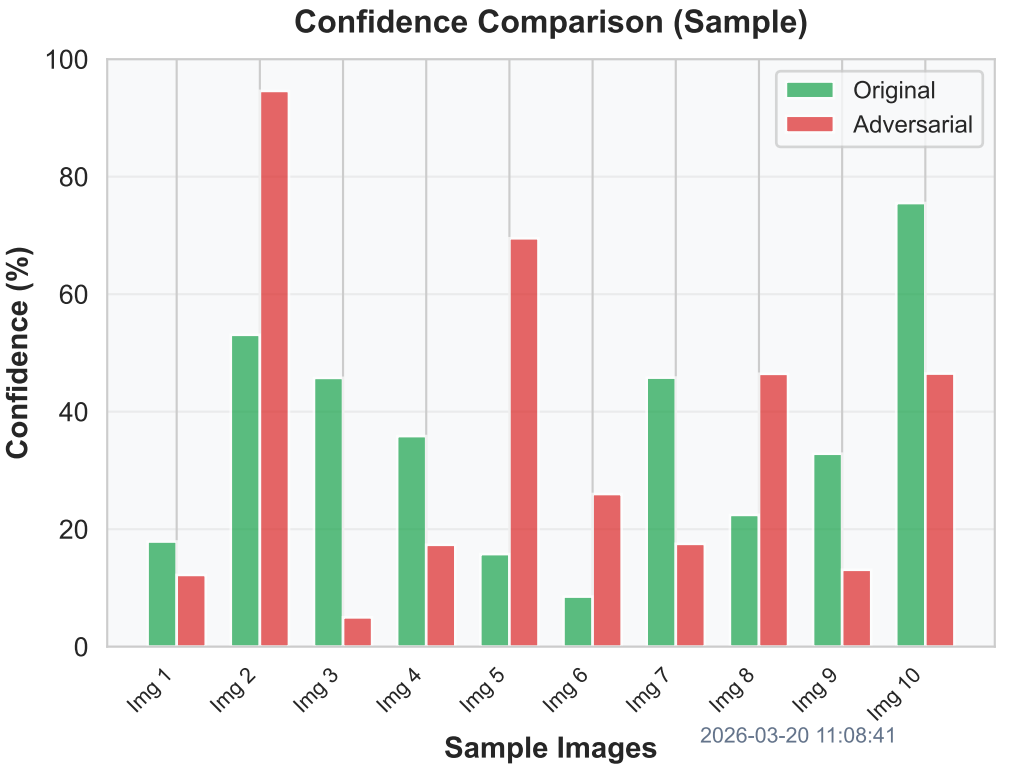
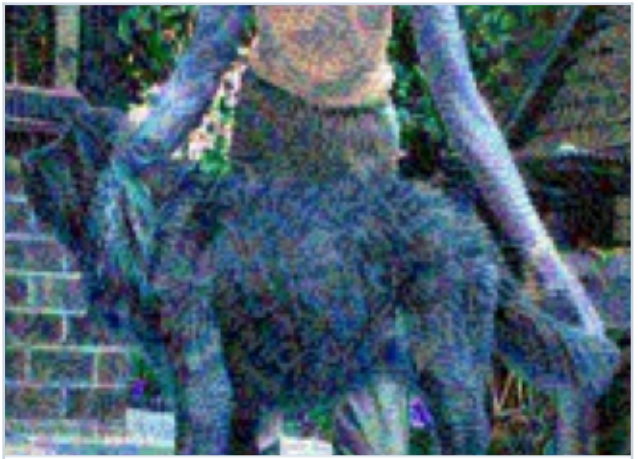


Image Evidence and Predictions

Images 1-4 of 10



Attack Successful
Image: Sussex_spaniel.JPEG
Outcome: Attack Successful
Original: Class 101
Confidence: 45.76%
Adversarial: Class 620
Confidence: 5.00%

Attack Successful
Image: flamingo.JPEG
Outcome: Attack Successful
Original: Class 815
Confidence: 22.43%
Adversarial: Class 620
Confidence: 46.45%

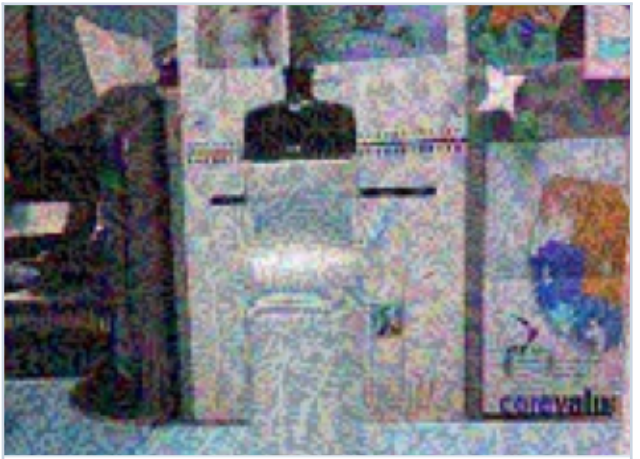
Attack Successful
Image: Dutch_oven.JPEG
Outcome: Attack Successful
Original: Class 469
Confidence: 45.80%
Adversarial: Class 463
Confidence: 17.51%

Attack Successful
Image: Model_T.JPEG
Outcome: Attack Successful
Original: Class 423
Confidence: 75.52%
Adversarial: Class 549
Confidence: 46.49%

Image Evidence and Predictions

Images 5-8 of 10

Attack Successful
Image: brown_bear.JPEG
Outcome: Attack Successful
Original: Class 270
Confidence: 35.85%
Adversarial: Class 611
Confidence: 17.35%



Attack Successful
Image: water_bottle.JPEG
Outcome: Attack Successful
Original: Class 477
Confidence: 8.53%
Adversarial: Class 549
Confidence: 25.99%

Attack Successful
Image: breastplate.JPEG
Outcome: Attack Successful
Original: Class 739
Confidence: 15.76%
Adversarial: Class 549
Confidence: 69.52%

Attack Successful
Image: hyena.JPEG
Outcome: Attack Successful
Original: Class 272
Confidence: 32.84%
Adversarial: Class 101
Confidence: 13.09%

Image Evidence and Predictions

Images 9-10 of 10

Attack Successful
Image: stage.JPEG
Outcome: Attack Successful
Original: Class 461
Confidence: 17.90%
Adversarial: Class 549
Confidence: 12.22%

Attack Successful
Image: jacamar.JPEG
Outcome: Attack Successful
Original: Class 92
Confidence: 53.09%
Adversarial: Class 94
Confidence: 94.60%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 35.35%, Average adversarial accuracy: 34.82%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques